

Farm Management System

Progress Summary & How Things Work

Date: 30 July 2026 | Stack: NestJS + Prisma + React (Vite) + PostgreSQL

Focus: Category / age batch tracking + monthly herd reports

1. What has been done

The farm ERP already had the 12 web modules (dashboard, animals, poultry groups, fish, expenses, revenue, P&L, inventory, health, breeding, production, reports). The latest completed feature replaces livestock-first individual listing with named category/age batches for livestock, poultry, and fish - with illness and mortality logging and downloadable monthly CSV reports.

1.1 Platform foundation (already in place)

- Monorepo: apps/api, apps/web, packages/contracts, packages/design-tokens
- Auth: JWT login + refresh; roles Admin / Manager / Worker with RBAC
- Local PostgreSQL + Prisma migrations and seed users
- Web UI: forest-green farm look, English + Nepali (en/ne)
- Modules for money (expenses approval, revenue, P&L) and operations (inventory, health, production)

1.2 Category / age batch tracking (this pass - complete)

- Prisma models: HerdBatch, BatchIllnessEvent, BatchMortalityEvent (+ migration)
- Unified batch kinds: LIVESTOCK | POULTRY | FISH with category, age range, counts
- Nest batches API: CRUD, illness log, mortality log, filters by kind/category
- Dashboard headcounts from sum of batch currentCount (by kind:category)
- Monthly herd report JSON + CSV (year, month, optional kind)
- Seed: example batches + illness/mortality; 1-2 breeding animals kept
- Web: /animals, /groups, /fish use shared batch cards; /batches/:id for detail
- Individuals demoted to Breeding stock at /animals/stock
- Reports UI: month/kind picker + Download monthly CSV
- Shared Zod schemas in @farm/contracts + en/ne copy for batches/reports

1.3 Out of scope (not done this pass)

- Photo marketplace-style cards for individual animals
- Changes to expense / revenue / P&L business logic

2. Domain model

Primary unit of tracking is a named HerdBatch (e.g. "Buffalo 2024 calves", age 0-12 months). Illness and deaths are count-based on the batch, not on named animals. Optional Animal records remain only for breeding parents.

```
HerdBatch
-----
kind          LIVESTOCK | POULTRY | FISH
category      e.g. BUFFALO, LAYER, ROHU
name          user label
ageFrom/ToMonths optional age band
initialCount  starting headcount
currentCount  live headcount now
deadCount     cumulative deaths

BatchIllnessEvent    condition + count + date
BatchMortalityEvent  count + reason + date --> reduces currentCount

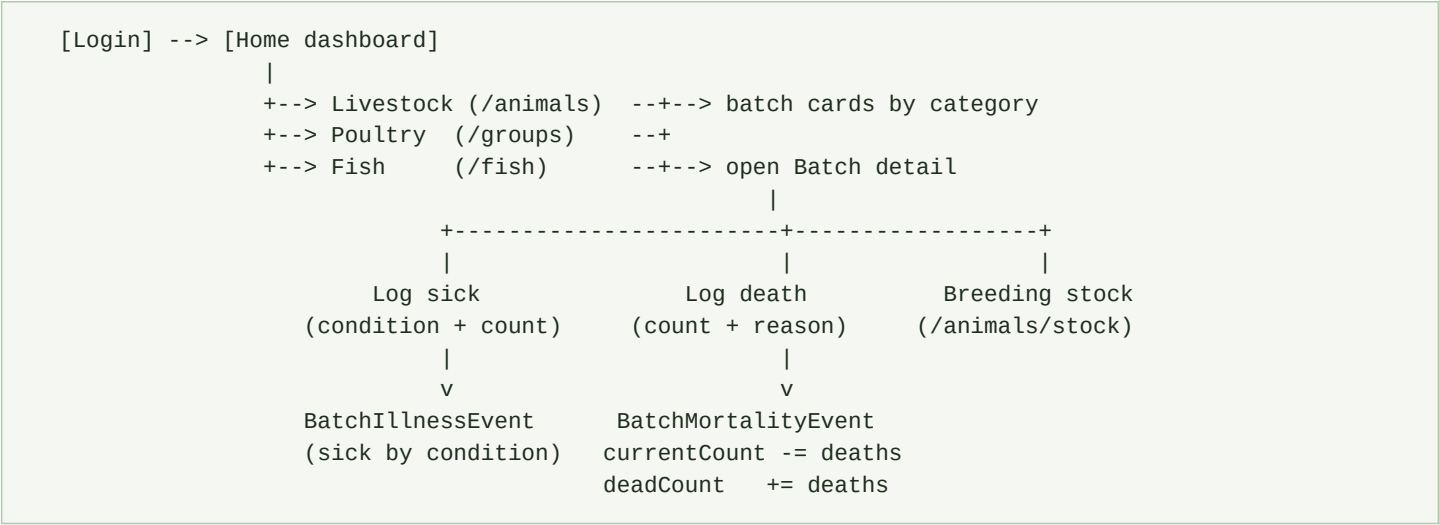
Animal (optional) --> BreedingRecord.motherId (breeding parents only)
```

Illness conditions (fixed vocabulary)

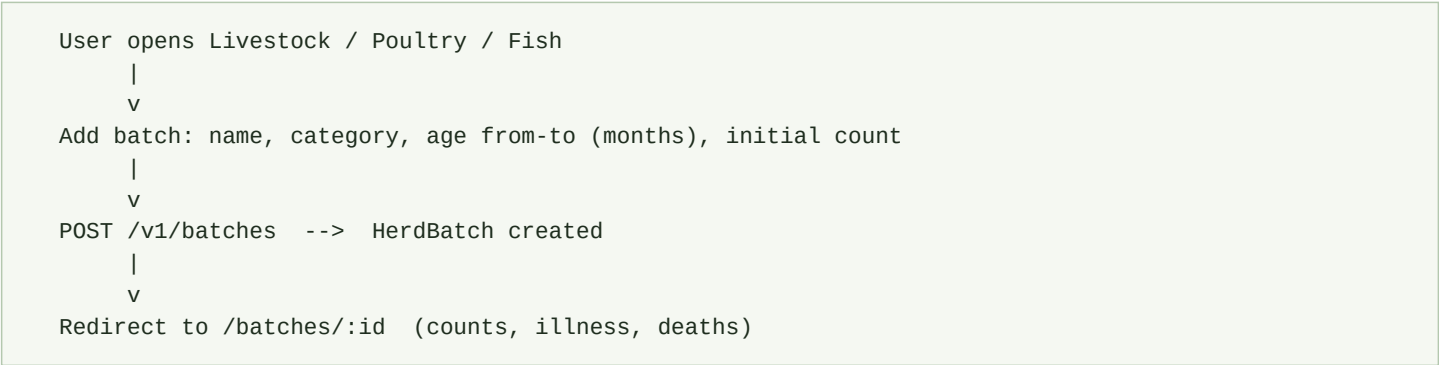
FMD, Mastitis, Newcastle, Parasites, Injury, Respiratory, Digestive, Skin, Other (extensible as string values in practice via the shared list in contracts).

3. How things work - workflows

3.1 Daily / ops workflow (batches)



3.2 Create a new batch



3.3 Monthly report workflow



3.4 Breeding (optional individuals)

```
Breeding page
|
+--> copy explains parents are optional tagged animals
|
+--> Manage breeding stock --> /animals/stock
|                               (list / add / edit Animal)
|
+--> Add mating: select mother (FEMALE animals), type, dates
                        --> BreedingRecord linked via motherId
```

Note: headcount for the farm comes from batches, not from Animal list size.

3.5 Dashboard snapshot

```
GET dashboard summary
|
+--> animalCount = SUM(HerdBatch.currentCount)
+--> speciesDistribution = group by "kind:category"
+--> alerts: health overdue, inventory critical, pending approvals
+--> finance tiles (if role can finance:read)
```

3.6 Request path (technical)

```
Browser (Vite :5173)
| proxy /v1
v
Nest API (:4000) -- JWT RBAC --> Batches / Reports / Dashboard modules
|
v
Prisma --> PostgreSQL (farm DB)

Shared types & Zod DTOs live in packages/contracts (@farm/contracts).
```

4. Key screens & routes

Route	Purpose
/dashboard	Home snapshot (batch headcounts)
/animals	Livestock batches (primary)
/animals/stock	Breeding stock (optional individuals)
/groups	Poultry batches
/fish	Fish batches
/batches/:id	Batch detail - sick / death logs
/breeding	Mating records + link to stock
/reports	Monthly herd report + other summaries
/expenses, /revenue, /pnl	Money modules (unchanged this pass)
/inventory, /health, /production	Ops modules (unchanged this pass)

Main API endpoints (batches / reports)

- GET/POST /v1/batches - list / create
- GET/PATCH/DELETE /v1/batches/:id
- GET/POST /v1/batches/:id/illness
- GET/POST /v1/batches/:id/mortality
- GET /v1/reports/herd-monthly
- GET /v1/reports/herd-monthly.csv

5. How to run locally

```
pnpm install
pnpm --filter @farm/contracts build
# PostgreSQL running; apps/api/.env configured
pnpm --filter @farm/api db:migrate
pnpm --filter @farm/api db:seed
pnpm dev:api      # http://localhost:4000
pnpm dev:web      # http://localhost:5173
```

Login: admin@farm.local / ChangeMe123!

Suggested next checks

- Create a livestock batch, log sick and a death, confirm counts update
- Download monthly CSV for current month
- Open Breeding stock and confirm it is separate from Livestock batches
- Switch language to Nepali and spot-check batch/report labels

This document summarizes work completed through the Category / age batch tracking pass. It does not replace docs/architecture.md, docs/rbac.md, or the OpenAPI docs at /docs.